RF Fingerprinting for Physical Layer Authentication: From Machine Learning to Deep Learning

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ABSTRACT Physical Layer Authentication (PLA) exploits device-specific hardware imperfections embedded in RF transient emissions to authenticate wireless nodes without cryptographic overhead. This report investigates the feasibility of PLA on a 12-device dataset using a deep learning pipeline comprising MLP, GRU, and Bidirectional GRU architectures. The project is based on a classical machine learning baseline (an SVM pipeline using RF-DNA features) which we rigorously re-evaluate to identify and eliminate three evaluation biases. Under such an evaluation protocol, both ML and DL models achieve an Average Detection Rate (ADR) in the 0.57–0.68 range, confirming that RF transient signals carry device-discriminating information. Sliding-window augmentation is shown to be a viable strategy for training DL models with fewer recordings, and a multi-class probe demonstrates that a single BiGRU trained on all 12 devices simultaneously outperforms per-device binary classifiers, pointing at a dataset size constraint rather than a model architecture one. These results establish a baseline for future work on larger datasets, adversarial evaluation, and system-level aggregation strategies.

I. INTRODUCTION

Physical Layer Authentication (PLA) leverages the unintentional hardware imperfections present in every radio transmitter to produce a device-specific RF fingerprint that is independent of the transmitted data and difficult to forge. Unlike cryptographic authentication, PLA operates at the signal level and requires no pre-shared secrets, making it an attractive complement to higher-layer security mechanisms in resource-constrained IoT deployments.

The power-on transient, the burst emitted by a device as it starts transmitting, is of particular interest for RF fingerprinting: hardware imperfections such as carrier frequency offset (CFO) and DC offset (DCO) imprint a device-specific signature on the IQ envelope during this transient. This signature is consistent across transmissions from the same device and is captured as a sequence of complex baseband (IQ) samples recorded by a software-defined radio at 20 MHz.

ORIGINAL RESEARCH DIRECTION

This project initially aimed to substitute the original ML pipeline with a DL one and then evaluate the adversarial robustness of the new PLA system: specifically, by implementing a white-box gradient attack on the trained classifiers and a black-box boundary one to assess how easily an adversary could craft RF signals that evade detection.

A structural obstacle motivated a change of direction: an adversarial analysis presupposes a reliable baseline, attacking a classifier whose honest accuracy is already near chance produces uninterpretable results, since there is no meaningful robustness to undermine.

REVISED CONTRIBUTION

The revised focus is on establishing a rigorous, reproducible performance baseline for RF fingerprinting on this dataset using both classical and deep learning pipelines. Specifically:

1) A common evaluation ground established by identifying and eliminating three methodological differences between the SVM and DL pipelines, including a data leakage and non-standard cross-validation protocols, to enable a fair comparison.
2) A complete deep learning pipeline — MLP on feature vectors, unidirectional GRU and Bidirectional GRU on raw DGT matrices — each evaluated under the same protocol across five random seeds and with leave-one-transient-out cross-validation.
3) A multi-class probe demonstrating that a single BiGRU trained on all 12 devices simultaneously outperforms per-device binary classifiers, pointing at a dataset size constraint rather than a model architecture one.
4) A data-efficiency result showing that sliding-window augmentation over the full transient duration is a viable strategy for training DL models with fewer recordings. Device-discriminating hardware imperfections (CFO, DCO) persist throughout the burst, so any 300-sample

window yields a valid fingerprint. Extracting 10 subsequences per transient grows the training set 10$\times$ (134 $\rightarrow$ 1340 samples) without additional recordings, making DL training feasible on a dataset that would otherwise be too small.

Taken together, these results confirm that RF transient emissions carry sufficient device-discriminating information to support PLA above chance, while providing a reproducible baseline from which future work on larger datasets, adversarial evaluation, and system-level aggregation strategies can build.

II. SYSTEM MODEL AND PIPELINE

A. DATASET

The dataset consists of RF transient recordings from 12 commodity IEEE 802.11 devices captured at 20 MHz with a software-defined radio. Each recording is a raw IQ capture from which the power-on transient is extracted via threshold-based detection; transients span approximately 100 000 samples ($\approx$5 ms). Approximately 11–13 raw transients per device are available, yielding roughly 134 recordings in total. Devices are partitioned into authorized and rogue sets according to three predefined trial configurations.

TABLE 1. Trial configurations: authorized vs. rogue device sets.

Trial	Authorized	Rogue
trial_1	dev3, dev2, dev12, dev9	remaining 8
trial_2	dev10, dev6, dev12, dev3, dev11, dev1	remaining 6
trial_3	dev1, dev10, dev9, dev8, dev12, dev11, dev6, dev7	remaining 4

B. FEATURE REPRESENTATIONS

Two complementary representations are extracted from each raw transient.

Feature Vector (FV). The DGT time-frequency matrix is divided into spatial patches and five statistics (mean, variance, skewness, kurtosis, range) are computed per patch, yielding a 505-dimensional feature vector per transient.

DGT Matrix. The raw 150$\times$150 Discrete Gabor Transform matrix is used directly, preserving the full time-frequency structure for sequential models.

C. WINDOWED AUGMENTATION

To increase effective sample count, 10 sub-sequences are extracted from each raw transient by distributing window start points evenly across the transient duration (via `np.linspace`), producing approximately 1340 samples per cache (FV and DGT). Each window is 300 IQ samples wide; because transients span roughly 100 000 samples at 20 MHz, consecutive windows are separated by $\approx$11 000 samples and do not overlap. The split is transient-aware: all windows derived from the same transient are assigned to the same partition, preventing any window-level leakage across train, validation, and test sets.

D. DEEP LEARNING ARCHITECTURES

Three architectures are evaluated on the binary classification task.

MLP-FV. A three-layer fully connected network ($505 \rightarrow 256 \rightarrow 128 \rightarrow 2$) with BatchNorm, ReLU, and Dropout ($p = 0.3$), operating directly on the 505-dimensional feature vector (163,458 parameters).

GRU-DGT. A 2-layer unidirectional GRU (hidden size 64) that reads the DGT matrix row-by-row as a sequence of 150 time frames, each with 150 frequency bins. The final hidden state feeds a linear classifier (66,562 parameters).

BiGRU-DGT + TF-Aug. A 2-layer bidirectional GRU (hidden size 64 per direction) that concatenates forward and backward final hidden states before classification (157,698 parameters). The small dataset and high intra-transient correlation from sliding-window augmentation would cause the BiGRU to overfit rapidly without regularisation. A custom time-frequency masking module counteracts this by randomly zeroing contiguous time rows, frequency columns, and adding Gaussian noise to the DGT matrix during training ($T_{\max} = 30$, $F_{\max} = 30$, $\sigma = 0.02$, $p = 0.5$), following the SpecAugment strategy.

All models are trained with AdamW (10^{-3}, wd $=10^{-4}$), batch size 64, and early stopping (patience 10) on validation loss with best-weight restoration.

E. CLASSIFICATION SCHEME

The system uses a binary one-vs-rest scheme: one classifier per authorized device is trained to distinguish that device (positive class) from all other authorized devices (negative class). Rogue devices are withheld from training entirely; they are only presented to the already-trained models at evaluation time to measure rejection performance. This training protocol is consistent with the original RF-DNA pipeline (provided as the course baseline), which likewise trains exclusively on authorized devices and treats rogue devices as unseen at training time. At inference time, a signal is accepted if, after claiming its identity, the corresponding classifier accepts it.

Performance is reported as:

- **Auth TVR**: True Accept Rate for the authorized device.
- **Rogue TVR**: True Reject Rate across all rogue devices.
- **ADR**: Average Detection Rate $=$ (Auth TVR $+$ Rogue TVR)$/2$.
- **FAR**: Probability a binary classifier accepts a rogue device.
- **FRR**: Probability a binary classifier rejects its authorized device.

F. EVALUATION PROTOCOL

Each model is trained under a group-aware 70/15/15 split: all windows from the same raw transient are assigned to the same partition, preventing window-level leakage. Z-score normalisation is fitted on the training fold only. A first pass reports mean $\pm$ std ADR across five random seeds. Because five seeds produce high variance for the BiGRU, Leave-One-Transient-Out (LOTO) cross-validation is used as the primary protocol: each fold holds out one complete transient as test, trains on the remaining, and selects the decision threshold on a validation split. The Equal Error Rate (EER) is computed from LOTO ROC curves at the operating point where $|\mathrm{FAR} - \mathrm{FRR}|$ is minimised.

III. RESULTS

A. ML REVISED EVALUATION

Three evaluation assumptions in the original ML pipeline inflate the reported ADR from ≈ 0.574 to near-perfect 0.973 when all corrections are removed (Fig. 1). The corrected SVM (Fix 1+2+3 with balanced class weights) forms the revised ML baseline for all subsequent comparisons. Full methodology and per-revision breakdown are given in Appendix.

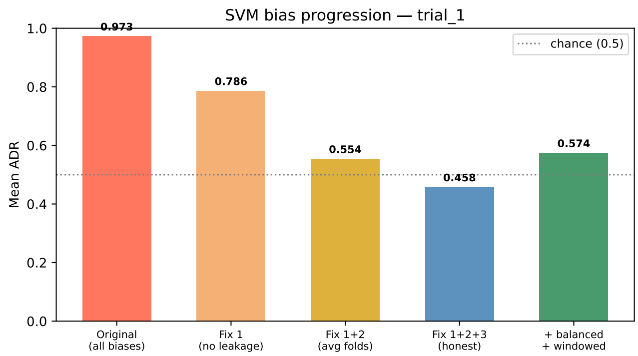

FIGURE 1. SVM ADR across pipeline variants (trial_1). Each correction removes one source of optimistic assumption; the rightmost bar is the revised baseline.

B. INITIAL DL EVALUATION (5 SEEDS)

Table 2 reports ADR for each DL model under a group-aware 70/15/15 split, averaged over five random seeds (trial_1). All three models achieve ADR between 0.62 and 0.68.

TABLE 2. DL model ADR on held-out test set (trial_1, 5 seeds).

Model	ADR (mean $\pm$ std)
MLP-FV	0.680 ± 0.027
GRU-DGT	0.623 ± 0.059
BiGRU+TF-Aug	0.645 ± 0.117

The BiGRU in particular exhibits high variance (seed ADR range 0.471–0.838). With only $n = 5$ seeds the confidence intervals are too wide to draw reliable conclusions about relative model performance. This motivated adopting LOTO cross-validation as the primary evaluation protocol, maximising use of the limited dataset.

C. LOTO CROSS-VALIDATION (PRIMARY EVALUATION)

LOTO (Leave-One-Transient-Out) holds out one complete transient per fold, making use of every available sample while

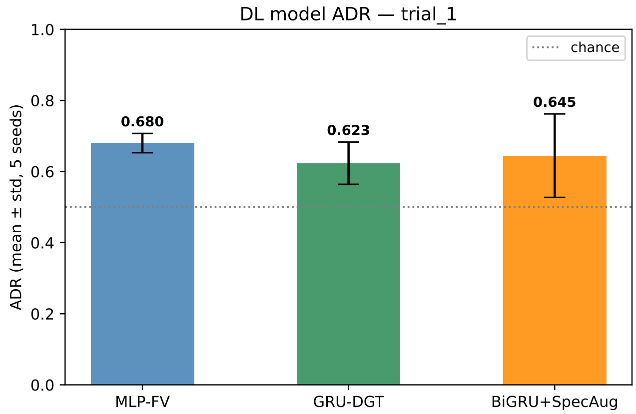

FIGURE 2. DL model ADR (mean $\pm$ std, 5 seeds, trial_1). The wide error bar on BiGRU signals that five seeds are insufficient for reliable conclusions.

remaining strictly transient-aware. Table 3 reports per-device ADR and EER for the BiGRU on trial_1. Mean ADR is 0.676 and mean EER is 0.262, well above the chance EER of 0.5, confirming above-chance device discrimination across all authorized devices.

TABLE 3. BiGRU LOTO per-device ADR and EER (trial_1).

Device	ADR	EER
device2	0.680	0.354
device3	0.569	0.299
device9	0.705	0.094
device12	0.748	0.302
Mean	**0.676**	**0.262**

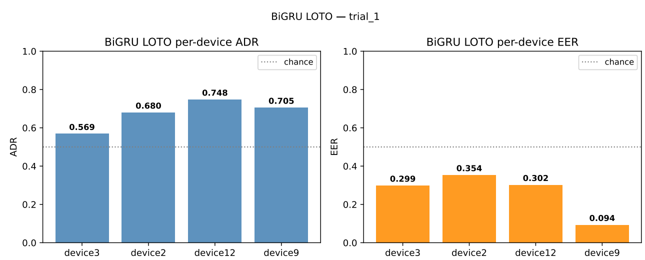

FIGURE 3. BiGRU LOTO per-device ADR and EER (trial_1). Dashed line marks chance level (EER $= 0.5$).

D. ML VS. DL COMPARISON

Fig. 4 places the revised SVM alongside all three DL models evaluated under LOTO cross-validation. No pipeline has a clear advantage: all models cluster in a 0.574–0.676 ADR band with overlapping uncertainty. The performance gap between the original and revised pipelines is entirely attributable to evaluation methodology.

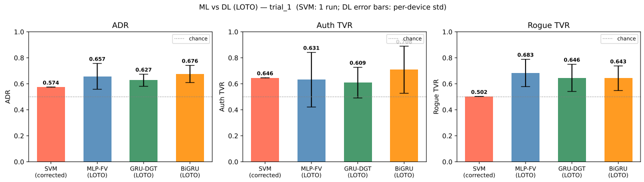

FIGURE 4. ML vs. DL (LOTO): ADR, Auth TVR, and Rogue TVR (trial_1). SVM error bar is zero (single deterministic run); DL bars are ± 1 std across authorized devices (LOTO).

IV. SUMMARY AND CONTRIBUTIONS

WHAT WE FOUND

- The original RF-DNA SVM pipeline contains three independent evaluation biases that collectively inflate reported ADR from $0.973 \to 0.574$ (approximately 40 percentage points).
- Under a fair evaluation protocol, SVM and all three DL models (MLP-FV, GRU-DGT, BiGRU+TF-Aug) achieve statistically equivalent ADR in the 0.57–0.68 range. No architecture is a clear winner.
- With only five seeds the BiGRU shows ADR variance of ± 0.131; LOTO cross-validation provides more stable per-device estimates (mean ADR 0.676, mean EER 0.262).
- Dataset size is the primary performance bottleneck: each binary classifier in trial_1 trains on $\approx$52 windowed samples, far below what DL models typically require to generalise reliably.

CONTRIBUTIONS

- **Revised baseline.** Identified, isolated, and quantified three evaluation assumptions in the original RF-DNA pipeline, providing reproducible corrections with per-bias ADR measurements.
- **Honest DL baseline.** First evaluation of MLP, GRU, and BiGRU architectures on this dataset under a strictly group-aware, transient-aware protocol across five seeds and LOTO.
- **Data-efficiency result.** Demonstrated that sliding-window augmentation enables DL training with $10\times$ fewer recordings by exploiting the full transient duration, confirming that device-discriminating hardware imperfections persist well beyond the initial signal segment used by the original pipeline.
- **Multi-class probe.** Showed that a single BiGRU trained on all 12 devices reaches 0.404 accuracy (vs. 0.083 random), confirming that discriminative information exists across all devices and that architecture is not the limiting factor while dataset size is (Appendix).

FUTURE DIRECTIONS

- Collect additional independent transients per device to push ADR toward operationally useful levels.
- Evaluate AND-aggregation at system level to reduce the high False Accept Rate under OR-aggregation.
- Explore per-device classifier ensembles: training multiple models per authorized device and combining their outputs via soft voting could reduce the high seed-to-seed variance observed in this work and yield more stable authentication decisions without architectural changes.
- Conduct adversarial robustness evaluation (white-box and black-box attacks on raw IQ waveforms), now that an honest unattacked baseline is established.

V. CONCLUSION

This report established a rigorous, reproducible performance baseline for RF fingerprinting on a 12-device dataset. A complete DL pipeline was evaluated under a rigorous, group-aware protocol; all models achieve ADR in the 0.57–0.68 range, confirming that RF transient emissions carry sufficient device-discriminating information to support PLA above chance, while highlighting per-classifier sample count as the binding constraint. Addressing this — through larger datasets or improved augmentation strategies — is the most direct path toward operationally useful PLA. Future directions are listed in Section IV.

Two results have direct implications for practical deployment. First, the high seed-to-seed variance of the BiGRU (ΔADR up to 0.37 across seeds for a single trial) means the system is not yet reliable enough for deployment at fixed thresholds; an ensemble or calibration step is needed before per-device classifiers can operate at a stable operating point. Second, the multi-class probe — which achieves substantially higher relative gain over chance despite identical data — suggests that joint training across all devices is the right inductive bias for a dataset of this size, and warrants prioritisation over further tuning of per-device binary classifiers.

APPENDIX. ML BIAS ANALYSIS

Fig. 1 shows the ADR progression across the four SVM pipeline variants for trial_1. The original biased pipeline reports near-perfect ADR, while each successive revision reduces performance toward the final level. The revised baseline (Fix 1+2+3 with balanced weights on windowed data) yields ADR ≈ 0.574 for trial_1.

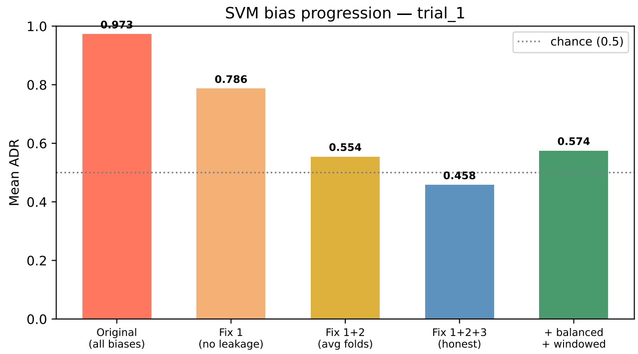

FIGURE 5. SVM ADR across pipeline variants (trial_1). Each revision removes one source of optimistic bias. The revised pipeline (Fix 1+2+3 + balanced, windowed) represents the honest ML baseline. The rightmost column shows the effect of windowed augmentation on the revised SVM pipeline. Group-aware cross-validation ensures no window-level leakage.

The revised SVM results confirm that the signal contains device-discriminating information — all devices achieve ADR above the chance level of 0.5 — but the reported performance is far lower than the original inflated figures suggested.

APPENDIX. MULTI-CLASS FEASIBILITY PROBE

To test whether the binary per-device setup is the binding constraint, a single BiGRU + TF-Aug is trained on all 12 devices simultaneously (12-class softmax). The model has access to the full 1340-sample windowed DGT cache, compared to the $\approx$52 samples available to each binary classifier in trial_1.

TABLE 4. Multi-class BiGRU vs. binary LOTO (trial_1, 5 seeds).

Setting	Metric	Value
Binary LOTO (BiGRU)	mean ADR	0.676
Multi-class (BiGRU)	accuracy	0.404 ± 0.057
Multi-class (BiGRU)	macro F1	0.323 ± 0.035
Random baseline (1/12)	accuracy	0.083

The multi-class BiGRU achieves 0.404% accuracy against a random baseline of 0.083, confirming that the RF transient carries device-discriminating information across all 12 devices. In absolute terms this is lower than the binary LOTO ADR (0.676), which reflects the harder 12-way classification task rather than a failure of the architecture. Devices 10–12, which have only 60 windowed samples each compared to 120–130 for devices 1–9, display the highest variance, reinforcing that per-device sample count is the primary performance driver.

APPENDIX. EXTENDED DISCUSSION

A. SCOPE OF THE REVISED AUDIT: WHY SVM AND ANOVA

The ML analysis in this work is deliberately scoped to the original RF-DNA pipeline from `RF_Fingerprint.py`: an SVM with polynomial kernel preceded by ANOVA-based feature selection (`SelectKBest` with the F-test statistic). This is not the only possible ML pipeline for the 505-dimensional feature vector. In fact the same code base also implements mutual information, RFE, and PCA as alternative selectors, and other classifiers such as Random Forest or k-NN could in principle be substituted. But auditing the specific choices made in the original work is the prerequisite for any fair comparison.

Why SVM. An SVM with polynomial kernel is a well-established choice for high-dimensional, small-sample classification problems. With $\approx$134 samples and 505 features, the dataset is firmly in the regime where SVMs with regularisation tend to outperform neural networks: there is not enough data to fit a network reliably, but the margin-maximising objective of the SVM generalises well when classes are linearly or polynomially separable in the feature space.

Why ANOVA. ANOVA F-test feature selection (`SelectKBest(f` `k=k))` ranks features by their univariate linear correlation with the class label. It is computationally cheap (no cross-validation loop required), has a single hyperparameter (k, the number of features to retain), and is easy to interpret. These properties make it a natural choice when the dataset is small and compute budget is limited.

Time constraints. Extending the revised audit to alternative classifiers and feature selectors was outside the scope of this project. Each additional pipeline variant requires its own replication, its own revision characterisation, and its own corrected baseline. Given the available time, the decision was to perform a thorough audit of the single pipeline that was actually used, rather than a shallow survey of many alternatives.

B. FEASIBILITY CONFIRMED, BUT OPERATIONALLY INSUFFICIENT

All pipelines achieve ADR consistently above the chance level of 0.5, which confirms the central premise of RF fingerprinting: hardware imperfections embedded in power-on transients are statistically discriminable by both classical and deep models. However, the gap between feasibility and operational deployment deserves careful framing.

C. MLP-FV VS. BIGRU: RESULTS DEPEND ON THE EVALUATION PROTOCOL

Comparing MLP-FV and BiGRU+TF-Aug requires reading both evaluations together, because the two protocols give different orderings.

In the simple held-out evaluation (one fixed 70/15/15 split per seed, five seeds), MLP-FV achieves a higher mean ADR (0.680 ± 0.027) than BiGRU+TF-Aug (0.645 ± 0.117). However, BiGRU's per-seed ADRs span from 0.471 to 0.838 — a range of 0.367 across just five runs — while MLP's range is only 0.065 (0.645–0.711). With such wide and overlapping confidence intervals ($n = 5$), the difference in means is not statistically significant. The higher mean of MLP in this evaluation is better explained by the fact that one BiGRU seed collapsed to ADR $= 0.471$ (below chance for rogue TVR) while another peaked at ADR $= 0.838$, averaging to a moderate figure that happens to fall below MLP's stable mean.

Under LOTO cross-validation — the more data-efficient and arguably more rigorous protocol, the ordering reverses: BiGRU achieves mean ADR $= 0.676$, MLP achieves 0.657, and GRU achieves 0.627. LOTO removes the threshold-variability issue (each fold calibrates its own threshold) and maximises training data, providing a more stable comparison surface. On this evidence, BiGRU is marginally better than MLP, not worse.

A structural reason why MLP-FV performs similarly to BiGRU is that its 505-dimensional feature vector is the result of deliberate domain engineering that distinguishes hardware imperfections. An MLP operating on this input does not need to learn which features matter because the engineering has done that; a BiGRU on raw DGT must discover the same information from scratch.

The practical implication is that the RF-DNA feature vector is not merely a legacy artefact: its domain-engineered statistics act as a built-in dimensionality reduction and noise suppression layer that compensates for the limited training set. This could serve as an intermediate representation within a deeper DL pipeline by pre-processing the signal into a compact, domain-informed descriptor and then feeding it to a more expressive classifier. This hybrid strategy could inherit the stability advantages of hand-crafted features while retaining the capacity to learn higher-level discriminative patterns, making it particularly well-suited to small-dataset RF fingerprinting regimes.

D. VALIDATION-BASED THRESHOLD CALIBRATION: TRIED AND REJECTED

A natural question when reporting LOTO results is whether the fixed decision threshold of 0.5 is optimal, or whether it can be improved by selecting the threshold from the available validation data. The LOTO pipeline was instrumented to answer this: after all folds complete, the validation scores from every fold are pooled per device and the threshold that maximises ADR on that pooled validation set is found via the ROC curve. This val-optimal threshold is then applied to the pooled test scores, giving a second set of results alongside the fixed-threshold results.

Table 5 shows the outcome for the BiGRU on trial_1.

TABLE 5. BiGRU LOTO: ADR at fixed threshold (0.5) vs. val-calibrated threshold (trial_1).

Device	ADR @ 0.5	ADR @ opt	Opt threshold
device2	0.680	0.624	0.303
device3	0.569	0.604	0.557
device9	0.705	0.710	0.851
device12	0.748	0.496	0.996
Mean	**0.676**	**0.608**	—

Val-calibration hurts overall: mean ADR drops from 0.676 to 0.608. The same pattern holds for MLP-FV (0.657 $\to$ 0.648) and GRU-DGT (0.627 $\to$ 0.605). The root cause is visible in the device12 row.

This is a direct consequence of the validation set being too small to reliably estimate a good threshold. Device12 has only 6 transients total; after reserving one as the test fold and one for early stopping per fold, the pooled val scores across all folds are too few and too concentrated to produce a stable operating point.

The practical conclusion is clear: **with this dataset size, the default threshold of 0.5 is more robust than any data-driven alternative**. Threshold calibration becomes worthwhile only when the calibration set is large enough to produce a smooth, representative score distribution...a condition this dataset does not meet.

E. MULTI-CLASS ACCURACY REINTERPRETED

At first glance, the multi-class BiGRU accuracy of 0.404 ± 0.057 appears lower than the binary LOTO ADR of 0.676, suggesting the multi-class model performs worse. This comparison is misleading because the two tasks differ fundamentally.

In binary LOTO, the model answers a yes/no question for one device against 8 rogues, with a balanced or threshold-corrected decision boundary. In multi-class, the model must identify the correct one among 12 devices. The appropriate reference is not 0.5 (binary chance) but $1/12 \approx 0.083$ (multi-class chance). Measured as relative improvement over the random baseline:

$$\text{Binary LOTO gain} = \frac{0.676 - 0.500}{0.500} = 35.1\%, \tag{1}$$

$$\text{Multi-class gain} = \frac{0.404 - 0.083}{0.083} = 386.5\%. \tag{2}$$

By this normalised measure, the multi-class model demonstrates more discriminative power, not less.

F. REPRESENTATIONAL LIMITS OF THE DGT APPROACH

The current pipeline extracts features from a single power-on transient. This is a one-time event spanning roughly 100 000 samples ($\approx$5 ms at 20 MHz): once the power-on transient ends, this fingerprinting opportunity is gone until the next power cycle. In real network scenarios, devices remain active for extended periods. The transient-only approach would require re-authentication at each power-cycle, which limits applicability in continuously running networks.

An alternative that has not been explored here is steady-state fingerprinting: exploiting clock imperfections, carrier frequency offset, or I/Q imbalance from normal data frames. Steady-state features can be extracted from any 802.11

packet, enabling continuous authentication without requiring special transient events. Combining transient and steady-state features would likely improve both accuracy and coverage.

Additionally, the DGT matrix of size 150×150 is large relative to the information it encodes per transient. Much of the matrix may be occupied by noise or uninformative regions. An attention mechanism (Transformer self-attention on the row sequence, or a spatial attention map over the 2D matrix) could learn to focus on the discriminative regions automatically, rather than requiring the entire matrix to be processed at equal cost.

G. MISSING EXPERIMENTS AND OPEN QUESTIONS

Several experiments that would substantially strengthen the conclusions were not performed, either due to dataset constraints or scope limitations.

TF-Aug ablation. As discussed above, the contribution of time-frequency masking augmentation to BiGRU performance is unknown because no BiGRU-without-TF-Aug baseline was recorded.

2D convolutional baseline. The GRU/BiGRU models impose a sequential scan over a 2D DGT matrix. A CNN applied to the same matrix as a 2D image would test whether the performance floor is architectural (sequential models are wrong for this data) or purely dataset-size related.

Adversarial evaluation. The original goal of the project was adversarial robustness evaluation. Now that a baseline exists, there could be a tractable attack direction.

Larger dataset evaluation. The most direct path to operational ADR is more data. This projection is speculative but motivates a data collection campaign as the first step toward a practical PLA system.

Which approach is more appropriate for this task?

Neither approach is clearly superior given the current dataset size. The SVM is more data-efficient and produces stable, reproducible results because it does not have random weight initialisation. However, its linear decision boundary in a fixed feature space is a fundamental capacity limit that cannot be overcome by adding more data.

Deep learning has no such architectural ceiling: given enough data, deeper or more complex architectures can fit arbitrarily complex decision boundaries. The multi-class probe already shows that the BiGRU, given access to $25\times$ more samples (1340 vs. 52), achieves a relative improvement over random chance that is $11\times$ larger than the binary SVM. This strongly suggests that deep learning is the right long-term direction, but that the current dataset does not yet have enough examples to let deep models demonstrate their advantage.